SAAS CUSTOMER CHURN INTELLIGENCE

One-page project report | Exploratory analysis of a 50,000-record SaaS customer dataset

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50,000	916	49,084	1.83%
Records analysed	Churned customers	Not churned	Overall churn share

Project objective and approach

The project explores customer churn patterns in a Software-as-a-Service (SaaS) business to identify groups that may need stronger retention attention. Using Python with pandas, Matplotlib and Seaborn, the notebook standardises column names, checks duplicate and missing values, converts date fields, summarises customer data, and segments churned customers by industry, subscription plan, country, payment status, account manager and activity dates.

Key findings

Overall churn: The notebook records 916 churned customers out of 50,000 records (about 1.83%); 49,084 are labelled not churned.

Plan tier: The Starter tier has the largest churn count (685), followed by Pro (174) and Enterprise (57). Starter customers are the clearest group for follow-up investigation by volume.

Industry: Among the five industries highlighted, EdTech has the highest churn count (128), followed by DevTools (121), Logistics (119) and MarTech (113); CyberSecurity is lowest (112).

Country: Singapore has the highest reported churned-customer count (135), followed by the USA (126) and France (125). The UK has the lowest count (95) among the countries listed.

Business recommendations

Prioritise a review of the Starter plan experience, including onboarding, product value, customer support and reasons for cancellation. Investigate EdTech churn and customer journeys in countries with higher counts, while treating counts as a starting point rather than proof of higher churn risk. Review payment status, feature adoption and account-manager patterns to identify possible churn signals and design targeted retention actions.

Limitations and next steps

The notebook mainly reports churn counts, not churn rates by segment. A larger count may reflect a larger customer base rather than a greater likelihood of churn. Calculate churn rate for each industry, plan tier and country using churned customers divided by all customers in that segment; then compare feature adoption, payment status and time since last activity. Validate data quality and patterns before making operational decisions.

Conclusion: The analysis establishes a useful baseline for SaaS churn monitoring. Segment-level churn rates and customer-behaviour analysis are the next steps toward evidence-based retention decisions.